U.S. DEPARTMENT OF ENERGY'S
CYBERFORCE® PROGRAM

CyberForce® 101

Security Documentation

January 2025

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CyberForce Competition® Overview

The CyberForce Competition® is an in-person competition that provides collegiate students with a cyber-physical hands-on, realistic energy-based scenario to secure and defend that incorporates life-like constraints and tasks, actual end users, and continual emerging threats. Students are challenged by the opportunity to apply classroom knowledge in a real-life practical environment.

The goal is to build the pipeline of qualified cybersecurity candidates for crucial positions such as those in our energy sector, national laboratories, and industry. The CyberForce Competition provides an inclusive space for all participants, including collegiate students, mentors, industry, and academia.

Security Documentation Overview

The **security documentation** is half of the *Orange Team Scoring* in the CyberForce Competition. It considers some of the soft-skills other portions of the competition do not have throughout the main day. Documentation is to be done utilizing the template provided and turned in prior to the competition.

The primary goal of the documentation section is to outline the provided infrastructure, vulnerabilities, changes made to the infrastructure to include security measures, and the reasonings why. Ideally the objective of this scoring section within the competition is to provide competitors with an opportunity to represent a small energy company that doesn't have documentation and to highlight the benefits and stress the

importance of needing to write clearly and thoroughly not just infrastructure changes but code and more in our daily work.

What is Technical Writing

Everyone benefits from a well-written document. Documentation at organizations can be used to educate personnel, record processes and policies, and report the results of audits and assessments. Ensuring that it is written clearly and thoroughly is essential to making clear decisions, action plans, or evaluating next steps.

The writing seen at many organizations is known as *technical writing*. This is “act of writing detailed information on a specific, often specialized topic”. Some examples include product manuals, user guides, internal processes and procedures, policies, compliance requirements, project plans, audits, and assessment documents.

Importance of Well-Written Documentation

Not only are you writing technical documentation, but it is imperative that the documentation is conveyed correctly.

- Write for the intended audience. Ensure that the audience is clearly understood prior to writing. For example, in the CyberForce Security Documentation, we articulate that the C-Suite would be reviewing the documentation. This means that both technical and non-technical people who are not typically engrossed in the day-to-day operations of this specific infrastructure will be looking at the security documentation. This means that acronyms should be spelled out, typical tech jargon should not be utilized, clarify meanings, and define terminology will improve the readability.

- **Be clear and concise**. The devil is in the details but too many details can cause confusion. Be sure that you are avoiding providing excessive and unnecessary information that is non-essential. If information feels non-essential but is required, consider an appendix or supplemental material to not lose the reader.
- **Do thorough research**. The documentation should reflect the expertise of the subject matter it presents. Fact-check details and document references. While there may be several "correct" ways to do something, ensure that you are documenting the way you want the reader to follow.
- **Write in stages**. If writing is not your strong suit, think about writing in phases. In the CyberForce writing, we provide a template which allows the writer(s) to go to areas that they are comfortable and work through and come back. No different to other technical writing, make an outline or template and write a rough draft. This allows you to identify gaps and reorganize.
- **Include Media Aids, if needed**. Graphics, charts, diagrams, etc., may be beneficial to help take difficult processes and make them simpler for people to follow. Use as needed but ensure they serve purpose.
- **Review and Revise**. There is nothing more frustrating than reading a document that has simple misspellings or grammatical errors that spell-check or someone else doing a quick read could have found. Ensure you provide some time to review and revise your document and abide by any constraints provided. If the requirement was 2 pages or 250 words, then ensure you are abiding by that.

Resources

- https://blog.rsisecurity.com/your-guide-to-writing-technical-documentation/
- https://www.giac.org/paper/gsec/4382/technical-writer-039-s-guide-documentation-information-security-arena/107250